



University of Aden -Faculty of Engineering
Information tech. Engg. Dept.



Logic design

Name :

Kamal Muthanna Ahmed Obaid

B2IT

Reg no. :

2405071028

Q) Design a logic circuit and function for the following:

Truth table:

binary	A	B	C	D	F
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	1
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

Solution

Using the Product of Sums (POS), we obtain the following:

binary	A	B	C	D	F
1	0	0	0	1	0
2	0	0	1	0	0
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0

$$F = M_1 \cdot M_2 \cdot M_6 \cdot M_7 \cdot M_8 = \prod(1,2,6,7,8)$$

The equation is:

$$(A+B+C+\bar{D}) \cdot (A+B+\bar{C}+D) \cdot (A+\bar{B}+\bar{C}+D) \cdot (A+\bar{B}+\bar{C}+\bar{D}) \cdot (\bar{A}+B+C+D)$$

Gate Design:

